Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **Java/Python services, React/JavaScript applications, and AI-powered workflows**; experienced in reusable architecture, data structures, algorithms, design patterns, testing, performance, and end-to-end product delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **scalable object-oriented Python services and reusable APIs** across FastAPI, PostgreSQL, SQL Server, and cloud storage, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Designed fault-tolerant AI workflow orchestration for 1,000+ jobs using explicit states, validation, and failure isolation, improving throughput 3x while sustaining 99%+ reliability across 5,000+ monthly production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **reusable React UI components and Python APIs** for uploads, dashboards, processing, history, and feedback, serving 150+ users and 10,000+ daily requests with stable production performance.
- Integrated OpenAI, Anthropic Claude, and Qwen into production workflows, owning features through requirements, design, implementation, test automation, code review, delivery, monitoring, and support while reducing turnaround 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **scalable object-oriented Java and Python application components** using Singleton and Factory patterns, REST APIs, SQL, caching, AWS Glue, and S3, improving response times 30% and maintainability.
- Collaborated with Microsoft and GitHub engineers through design discussions and code reviews while maintaining 20+ unit and integration tests, Git workflows, Jenkins builds, deployment automation, and documentation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **scalable React, JavaScript, and Node.js product features** across eight modular services for authentication, records, billing, permissions, and notifications, securely synchronizing 2M+ records through REST APIs and WebSockets.
- Enhanced application usability and performance through reusable components, caching, automated testing, and production observability, reducing resource waste 42% and deployment time 50% while achieving 95%+ build validation.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **scalable customer-facing React, TypeScript, and Python features** for booking, pricing, payments, refunds, and status workflows, supporting 50,000+ records through real-time synchronization with sub-200-millisecond API responses.
- Worked directly with stakeholders to clarify requirements and design 12+ integrations, implementing secure authentication, validation, and webhooks while reducing manual refund handling 70% across mobile and web products.

Skills

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, HTML5, CSS3
Frontend: React.js, Next.js, Angular.js, reusable components, responsive design, Material UI, TailwindCSS
Engineering: Data structures, algorithms, OOP, Singleton/Factory patterns, system design, performance optimization, scalability
AI & Backend: OpenAI, Anthropic Claude, Qwen, LLM orchestration, FastAPI, Flask, Node.js, REST APIs, PostgreSQL
Tools: Cursor, Git, GitHub Actions, Jenkins, Docker, Kubernetes, Jest, PyTest, CI/CD, debugging, code reviews, Unix

Projects

Big-Brain Live | *PostgreSQL, LLMs, REST APIs*

- Built an **AI-powered workflow application** processing 50,000+ documents with authenticated uploads, PostgreSQL metadata, incremental indexing, LLM summarization, semantic retrieval, REST APIs, and collaborative workflows, reducing review time 60%.

HMIS Live | *TypeScript, REST APIs, Sentry*

- Built an **extensible, customizable workflow product** with reusable interfaces, role-based access, dashboards, appointments, REST integrations, notifications, and monitoring, supporting 100+ daily workflows while reducing administrative workload 30%.

Chat2SQL Code | *Java, TypeScript, Docker, Maven*

- Enhanced an **existing cross-platform full-stack Java and TypeScript application** supporting 10+ databases with schema exploration, reusable interfaces, query generation, validation, Docker, Maven, Node.js, and structured debugging workflows.